Lesson 1: What Is a System of Linear Equations? Solving by Graphing

Two equations at once — the solution is the single point where both lines cross.

1. What Is a System of Linear Equations?

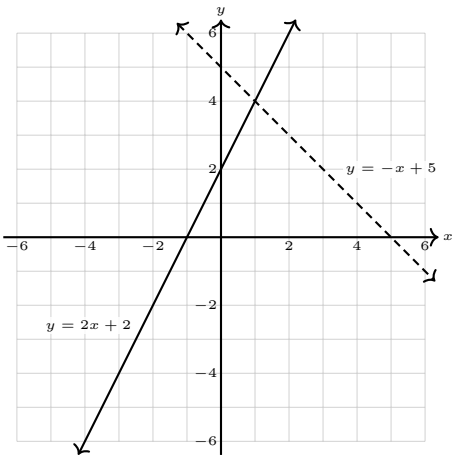
The equation $y = x + 2$ is true for many pairs: $(0, 2)$, $(1, 3)$, $(4, 6)$. A second equation, $y = -x + 6$, has many pairs of its own: $(0, 6)$, $(1, 5)$, $(6, 0)$. Only one pair makes *both* true at once — $x = 2$, $y = 4$, since $4 = 2 + 2$ ✓ and $4 = -2 + 6$ ✓. Two linear equations considered **together** form a **system of linear equations**, and the ordered pair (x, y) that satisfies **every** equation in the system is the **solution to the system**.

2. The Solution Is the Point of Intersection

Every point on a line satisfies that line's equation, so a point lying on **both** lines satisfies **both** equations. Graph the two equations on the same grid and the **point of intersection** — where the lines cross — is the solution to the system.

Example 1

The system $y = 2x + 2$ and $y = -x + 5$ is graphed below; the lines cross at integer coordinates.



(a) State the solution: $x =$_____, $y =$_____

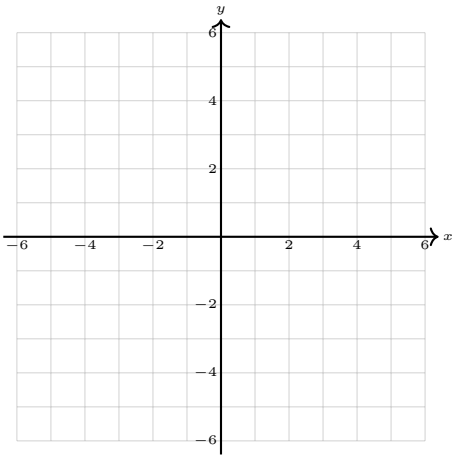
(b) Write the solution as an ordered pair.

3. Solving a System by Graphing

Step 1: if an equation is not already $y = mx + b$, rewrite it in slope-intercept form. (If it has $-y$, e.g. $2x - y = 4$: subtract $2x$ to get $-y = -2x + 4$, then change every sign: $y = 2x - 4$.) **Step 2:** graph both lines on the same grid using each y-intercept and slope. **Step 3:** find where they cross. **Step 4:** state that point as an ordered pair (x, y).

Example 2

Consider the system $y = x - 3$ and $y = -2x + 3$.

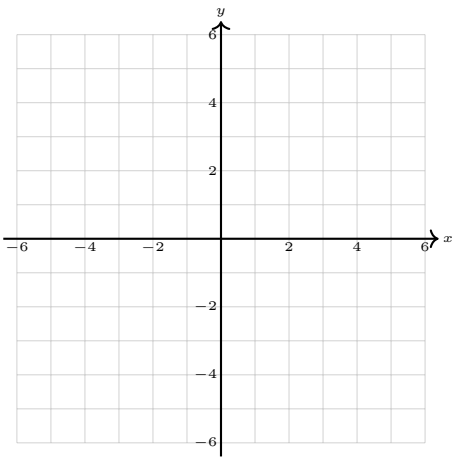


(a) Graph both equations on the grid at left, without using technology.

(b) State the solution to the system as an ordered pair.

Example 3

Consider the system $2x + y = 1$ and $y = x - 5$.



(a) Rewrite $2x + y = 1$ in slope-intercept form.

(b) Graph both equations on the grid at left, without using technology.
(c) State the solution as an ordered pair.

4. Checking a Solution

A point read off a hand-drawn graph is easy to get slightly wrong, so substitute it into **both original equations**. If both come out true the solution is confirmed; if *either* one is false, the graph or the reading of it went wrong.

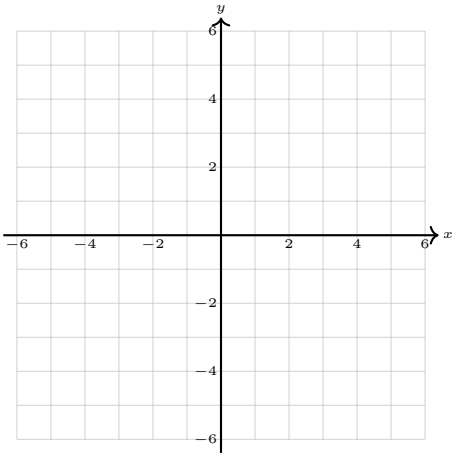
Example 4

Determine whether the given ordered pair is the solution to the system.
(a) $(3, 1)$ for $y = 2x - 5$ and $y = -x + 4$.

(b) $(2, 3)$ for $y = 3x - 4$ and $y = -2x + 9$.

Example 5

Consider the system $x + y = 3$ and $y = 2x - 3$.



(a) Rewrite $x + y = 3$ in slope-intercept form.

(b) Graph both equations on the grid at left.
(c) State the solution as an ordered pair.

(d) Check the solution in **both original** equations, $x + y = 3$ and $y = 2x - 3$.

Lesson 2: Classifying Systems: One Solution, No Solution, or Infinite Solutions

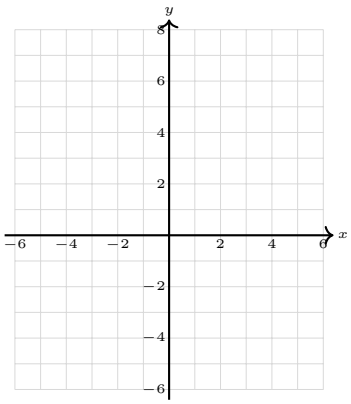
Two lines need not cross once — they may never cross, or lie exactly on top of each other.

1. One Solution — Different Slopes

Two lines with different slopes are tilted by different amounts, so wherever they start they must cross — and they can cross only **once**. The system has exactly **one solution**: the coordinates of that crossing point. This is by far the most common case.

Example 1

Consider the system $y = 2x - 1$ and $y = -x + 5$.



(a) State the slope of each line. Are the slopes the same or different?

(b) Graph both lines on the grid at left.

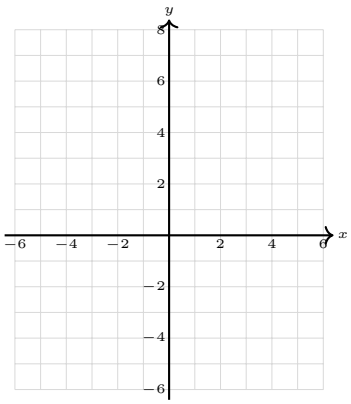
(c) State the point where the lines cross. This is the solution to the system.

2. No Solution — Parallel Lines

Two lines are **parallel** exactly when they have the same slope. Equations with the **same slope** but **different y-intercepts** graph as parallel, distinct lines that stay the same distance apart forever and never meet: the system has **no solution**.

Example 2

Consider the system $y = 3x + 2$ and $y = 3x - 3$.



(a) State the slope and y-intercept of each equation.

(b) Graph both lines on the grid at left. What do you notice?

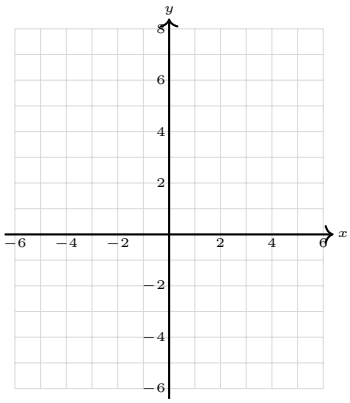
(c) How many solutions does this system have?

3. Infinitely Many Solutions — The Same Line in Disguise

Two equations that look different can be the **exact same line** — one is just a multiple of the other, so the slopes *and* the y-intercepts match. One graph lands on top of the other, so **every** point on the line is a solution: **infinitely many solutions**. To compare, rewrite each equation as $y = mx + b$, dividing *every* term by the number in front of y: $2x + 2y = 8 \Rightarrow 2y = -2x + 8 \Rightarrow y = -x + 4$.

Example 3

Consider the system $x + y = 4$ and $2x + 2y = 8$. Notice that $2x + 2y = 8$ is exactly $2 \times (x + y = 4)$.



(a) Rewrite both equations in slope y-intercept form.

(b) Graph both lines on the grid at left.

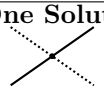
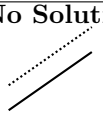
(c) How many solutions does the system have? Explain.

4. Classifying a System Without Graphing

Step 1: write both equations in slope y-intercept form, $y = mx + b$. **Step 2:** compare the slopes — **different** slopes means **one solution**, and you may stop. **Step 3:** if the slopes are the **same**, compare the y-intercepts: **different** intercepts means **no solution**, the **same** intercept means **infinitely many solutions**. Far faster than graphing.

Example 4

Use what you noticed in Examples 1–3 to complete the chart below.

One Solution	**No Solution**	**Infinitely Many**
 Slopes: _____ Intercepts: _____	 Slopes: _____ Intercepts: _____	 Slopes: _____ Intercepts: _____

Example 5

Without graphing, determine whether each system has one solution, no solution, or infinitely many solutions.

(a) $y = 4x + 3$, $y = 4x - 1$ (b) $y = 5x + 2$, $y = -3x + 2$ (c) $y = 2x - 6$, $y = 2x - 6$ (d) $y = -x + 8$, $y = x + 8$

Lesson 3: Solving Systems by Substitution, Part 1: When a Variable Is Already Isolated

Trading a variable for an equal expression to get an exact solution without graphing.

1. The Idea Behind Substitution

Graphing gives a solution only as accurate as the grid; substitution gives the **exact** answer every time. If one equation says $y = 2x + 1$, then "y" and "$2x + 1$" are two names for the same value, so wherever y appears in the **other** equation we may write $2x + 1$ instead. That other equation then holds only x — one equation, one unknown, which we already know how to solve.

2. The Substitution Process, Step by Step

Find the equation already written as "$y = \ldots$" or "$x = \ldots$"; substitute that expression into the *other* equation; solve the resulting one-variable equation; back-substitute that value into the isolated equation to get the second variable; then state the solution as an ordered pair (x, y) and check it in *both* originals. Substitute into the **other** equation — putting the expression back into the equation it came from collapses to a true-but-useless identity such as $2x + 1 = 2x + 1$.

Example 1

Solve $y = 2x + 1$ and $3x + y = 16$ by substitution, one step at a time.

Step 1: Which equation already has a variable isolated? Write it down.

Step 2: Substitute $2x + 1$ in place of y in $3x + y = 16$. The new equation should contain only x.

Step 3: Solve the equation from Step 2 for x.

Step 4: Substitute that x-value back into $y = 2x + 1$ to find y.

Step 5: Write the solution as an ordered pair (x, y), then check it in *both* original equations.

3. Practicing Substitution

Each system below already has one equation solved for a variable. Substitute, solve, back-substitute, and check. The isolated variable may be x rather than y, and a solution may come out negative — both are perfectly normal.

Example 2

Solve $y = x - 4$ and $2x + 3y = 18$ by substitution.

(a) Substitute $x - 4$ for y in the second equation and solve for x.

(b) Substitute your x-value back into $y = x - 4$ to find y.

(c) Check your solution in both original equations.

Example 3

Solve $x = 3y + 1$ and $x + 2y = 16$ by substitution.

(a) Substitute the expression for x into the second equation and solve the system.

(b) Check your solution in both original equations.

Example 4

Solve $y = -2x + 5$ and $3x - y = 10$ by substitution, and check. One variable will come out negative.

4. Checking the Solution

Check every ordered pair in **both** original equations — not the rearranged ones — and confirm each left side equals its right side. A check that does not balance catches sign errors and missed steps, and a pair that satisfies only one equation is not a solution to the system at all.

Example 5

Solve $y = 3x - 7$ and $5x - 2y = 11$ by substitution. Show the substitution step clearly, then check the solution in **both** original equations, writing each check as its own line of work.

Lesson 4: Solving Systems by Substitution, Part 2: Isolating a Variable First

When neither equation is solved for a variable, create that starting point yourself.

1. The Extra Step: Isolate Before You Substitute

In Lesson 3 one equation already read $y = \ldots$ or $x = \ldots$. In a system like $x + 2y = 11,\ 3x - y = 5$ neither variable stands alone, so substitution needs one new step first: **isolate a variable yourself**, then continue exactly as before. Choose well — scan both equations for a coefficient of **1** or $-$**1** and isolate *that* variable, since it takes one step and never creates a fraction. Here x in Equation 1 gives $x = 11 - 2y$ and y in Equation 2 gives $y = 3x - 5$; isolating the $2y$ or the $3x$ instead would force $\dfrac{11 - x}{2}$ or $\dfrac{5 + y}{3}$.

2. The Full Five-Step Process

Isolate a variable in one equation, **substitute** that expression into the *other* equation, **solve** for the remaining variable, **back-substitute** into the Step 1 expression, then **check** both values in the original equations and state the ordered pair. For $x + 2y = 11,\ 3x - y = 5$: $x = 11 - 2y$, so $3(11 - 2y) - y = 5$, giving $33 - 7y = 5$, $y = 4$; then $x = 11 - 2(4) = 3$. Both originals check, so the solution is $(3, 4)$.

Example 1

Solve the system $2x + y = 12,\ x - 4y = -3$ completely, showing all five steps.

Step 1 — Isolate a variable (look for a coefficient of 1 or -1).

Steps 2–3 — Substitute into the other equation, then solve it.

Steps 4–5 — Back-substitute, check both originals, and state the ordered pair.

Example 2

Solve the system $x + 5y = 17,\ 4x - y = 5$, working through the stages below.
(a) Steps 1–2: isolate a variable, then substitute into the other equation.
(b) Steps 3–5: solve, back-substitute, check, and state (x, y).

Example 3

For $5x + y = 21,\ 2x - 3y = 5$, only *one* variable in the whole system has a coefficient of 1 or -1.
(a) Which variable will you isolate, and in which equation? Why?
(b) Solve the system completely.

Example 4

Solve $3x - y = 8,\ x + 4y = 7$. The variable you isolate has a coefficient of -1 — watch your signs.

Example 5

Solve $2x + 3y = 14,\ 5x - 4y = 12$. *No* coefficient here is 1 or -1, so isolating creates a fraction; carry it carefully and it will cancel by the end.

Lesson 5: Solving Systems by Elimination, Part 1: When Coefficients Already Match

Adding or subtracting two equations to make one variable vanish before you solve.

1. The Idea Behind Elimination

Substitution always works, but isolating a variable can create fractions to drag through every step. **Elimination** lines the two equations up and combines them so one variable disappears entirely. If $5x + 2y = 16$ and $5x - 3y = 1$, subtracting gives $(5x + 2y) - (5x - 3y) = 16 - 1$, so $5y = 15$ and $y = 3$ — the x-terms cancelled because $5x - 5x = 0$, leaving one equation in one variable.

2. Eliminating by Subtraction (Matching Coefficients)

If a variable has the *exact same* coefficient — same number, **same sign** — in both equations, **subtract** the equations. Line up x-terms, y-terms and constants in columns and subtract each column. **Watch out:** subtracting changes the sign of *every* term of the equation you subtract, constant included — $23 - 11$, not $23 + 11$.

$$\begin{array}{r} 3x + 4y = 23 \\ -)\ 3x - 2y = 11 \\ \hline 6y = 12 \end{array}$$

Then solve the one-variable equation and back-substitute into either *original* equation.

Example 1

For the system $3x + 4y = 23$ and $3x - 2y = 11$, work through the elimination one step at a time.

(a) Do the x-terms have the same coefficient? Which variable does subtracting Eq. 2 from Eq. 1 remove?

(b) Subtract Equation 2 from Equation 1, lining up like terms. Write the resulting one-variable equation.

(c) Solve the equation from part (b) for y.

(d) Substitute your y into either original equation to find x, then check in *both* originals.

3. Eliminating by Addition (Opposite Coefficients)

Sometimes the coefficients are not equal but *opposite* — same number, **opposite signs**, like $+5y$ and $-5y$. Then **add** the two equations: left sides together, right sides together. A number plus its opposite is 0, so $5y + (-5y) = 0$ and the variable is gone. Solve, then back-substitute and check in both original equations, exactly as before.

Example 2

For the system $2x + 5y = 23$ and $6x - 5y = 9$, eliminate by adding.

(a) Are the y-terms opposites? Which variable does adding the two equations remove?

(b) Add the two equations. Write the resulting one-variable equation, then solve it for x.

(c) Substitute your x into either original equation to find y, then check both originals.

4. Deciding: Add or Subtract?

Before touching a new system, ask one question — *do a variable's coefficients match, or are they opposite?* Same number, **same** sign (e.g. $+4x$ and $+4x$) $\longrightarrow$ **subtract**. Same number, **opposite** sign (e.g. $+4x$ and $-4x$) $\longrightarrow$ **add**. Either way one variable disappears, leaving an equation you already know how to solve.

Example 3

For the system $4x + 3y = 18$ and $7x + 3y = 36$, decide, eliminate, and solve.

(a) Add or subtract? Look at the coefficients of y in both equations and decide.

(b) Eliminate y and solve the resulting equation for x.

(c) Back-substitute to find y, then check both original equations.

Example 4

For the system $4x + y = -7$ and $-4x + 3y = 27$, decide, eliminate, and solve.
(a) Add or subtract? Look at the coefficients of x in both equations and decide.

(b) Eliminate x and solve the resulting equation for y.

(c) Back-substitute to find x, then check both original equations.

Example 5

For $-2x + 5y = -34$ and $-2x - 3y = -2$, both coefficients of x are negative — watch your signs.
(a) Add or subtract? Look at the coefficients of x in both equations and decide.

(b) Eliminate x and solve the resulting equation for y.

(c) Back-substitute to find x, then check both original equations.

Lesson 6: Solving Systems by Elimination, Part 2: Multiplying First

Scaling one or both equations by a constant so a pair of coefficients finally matches.

1. When Neither Coefficient Matches

In Lesson 5 one variable always arrived with matching or opposite coefficients. In $x+2y = 9$ and $3x-y = 13$ the x-coefficients are 1 and 3, the y-coefficients are 2 and -1 — neither pair matches and neither pair is opposite, so adding or subtracting as written eliminates nothing. Elimination is still available; it just needs one extra step first: **multiply** an equation by a constant.

2. Multiplying One Equation

Look for a coefficient that is already a *factor* of the other. Here 1 divides into 2: multiplying Equation 2 by 2 turns $-y$ into $-2y$, the exact opposite of the $2y$ above it. **Multiply every term on both sides** — the commonest error is scaling the left side and forgetting the constant.

$$\begin{array}{r} x + 2y = \ 9 \\ +)\ 6x - 2y = 26 \\ \hline 7x = 35 \end{array}$$

Then solve, back-substitute into an *original* equation, and check the pair in both.

Example 1

For the system $x + 2y = 9$ and $3x - y = 13$, build the match one step at a time.

(a) Does adding or subtracting the equations as written eliminate either variable? Explain.

(b) The y-coefficient of Equation 2 is -1. By what constant should you multiply Equation 2 to make it the opposite of $2y$? Write the new Equation 2.

(c) Add Equation 1 to your new Equation 2 from part (b). Solve the resulting equation for x.

(d) Substitute your x into either *original* equation to find y, then verify in both originals.

3. Multiplying Both Equations

When neither coefficient divides the other — say 3 and -2 — **both** equations must be multiplied. Take the **lowest common multiple** of the two coefficients (3 and 2 give 6), then scale each equation so that variable becomes $+6y$ in one and $-6y$ in the other. Using the LCM rather than any common multiple keeps the numbers as small as possible.

Example 2

For the system $2x + 3y = 17$ and $3x - 2y = 6$, eliminate y by multiplying both equations.

(a) The y-coefficients are 3 and -2. Neither is a factor of the other; find the lowest common multiple of 3 and 2.

(b) Multiply Equation 1 by the constant that turns its y-coefficient into your answer to (a), and Equation 2 by the constant giving the *opposite* of it. Write both new equations.

(c) Add your two new equations from part (b). Solve the resulting equation for x.

(d) Substitute your x into either original equation to find y, then verify in both originals.

4. Choosing the Easier Variable

Scan *both* variables before you multiply anything. If one variable can be eliminated by multiplying just **one** equation, that route is faster and far less error-prone than one needing **both**. The full procedure: choose a variable, multiply one or both equations, add or subtract, solve, back-substitute, then state the solution as an ordered pair.

Example 3

For the system $5x - 2y = 14$ and $3x + 4y = 24$, compare the two routes before committing.

(a) For each variable, decide whether you would multiply just one equation or both. Which variable is easier to eliminate?

(b) Eliminate the easier variable (multiply, then add or subtract) and solve for x.

(c) Substitute to find y, then verify the solution in both original equations.

Example 4

For the system $3x + 4y = 29$ and $5x - 6y = -15$, both equations will need multiplying.

(a) The y-coefficients are 4 and -6. Find their lowest common multiple, and state the constant you would multiply *each* equation by to reach it.

(b) Multiply both equations as in part (a), then add or subtract to eliminate y. Solve for x.

(c) Substitute to find y, then verify the solution in both original equations.

Example 5

A bake sale sells cookies and cupcakes. Friday: 3 cookies and 5 cupcakes sold for \$19. Saturday: 4 cookies and 2 cupcakes sold for \$16.

(a) Let x be the price of a cookie and y the price of a cupcake, both in dollars. Write a system of two equations describing Friday's and Saturday's sales.

(b) Solve your system by elimination. (Both equations will need to be multiplied.)

(c) State the price of a cookie and the price of a cupcake, then verify your solution in both original equations.

Lesson 7: Solving Word Problems with Systems of Equations

Turning a paragraph about two unknown quantities into two equations, then solving it.

1. The Four-Stage Method

The algebra is nothing new — substitution and elimination already work. The new skill is **translating**, and every problem here runs the same four stages: **define the two variables** (say exactly what each letter counts, with units), **write two equations** — one from each fact in the problem, **solve** by substitution or elimination, then **answer in a sentence** and check that the answer is reasonable in context. Key words name the operation: *sum, total, altogether* and *more than* $\rightarrow +$; *difference, less than, fewer than* $\rightarrow -$; *twice, double, triple* $\rightarrow \times$. Watch the order: "6 less than $3y$" is $3y - 6$, not $6 - 3y$.

Walkthrough. *"Two numbers have a sum of* 22*; the larger is* 6 *more than the smaller."* Let L = larger, S = smaller, so $L + S = 22$ and $L = S + 6$. Substituting, $(S + 6) + S = 22$, giving $2S = 16$, $S = 8$, $L = 14$. The numbers are 14 and 8: $14 + 8 = 22$ ✓ and $14 = 8 + 6$ ✓.

Example 1

There are 30 students in Ms. Alvarez's classroom. There are 8 more girls than boys.

(a) Assign a variable to each unknown.

Let $b =$ _____

Let $g =$ _____

(b) Translate the two given facts into a system.

Equation 1: _____

Equation 2: _____

(c) Solve the system to find the number of boys and the number of girls.

(d) State the answer in words, and check both values in the original equations.

Answer: _____

2. Number and Age Word Problems

Two related numbers — often two people's ages — are tied together by a **sum** and by a **comparison** ("5 years older than", "3 times as old as"). The comparison is usually already isolated, so substitution is quickest.

Example 2

Jordan is 5 years older than his sister Ava. The sum of their ages is 27.

(a) Let J represent Jordan's age and A represent Ava's age. Write a system of two equations that models this situation.

Equation 1: _____ Equation 2: _____

(b) Solve the system using **substitution**.

(c) State both ages in words, and check your answer.

Answer: _____

3. Cost and Purchase Word Problems

Two different items sell at two different prices. One equation counts **how many** items there are altogether; the other totals their **value** — price times quantity for each item, added. Both come out in standard form, so elimination usually fits best.

Example 3

At the school canteen, hot dogs cost \$3 and juice boxes cost \$2. On Monday, the canteen sold a combined total of 40 hot dogs and juice boxes, for total sales of \$97.

(a) Let h represent the number of hot dogs sold and j represent the number of juice boxes sold. Write a system of two equations (one for the total number of items, one for the total sales).

Count equation: _____ Value equation: _____

(b) Solve the system using **elimination**.

(c) State the answer in words, and check.
Answer: _____

Example 4

At a school concert, student tickets cost \$5 and adult tickets cost \$8. A total of 50 tickets were sold, for total ticket revenue of \$316.

(a) Let s represent the number of student tickets sold and a represent the number of adult tickets sold. Write a system of two equations, then solve it using **elimination**.

(b) State the answer in words, and check.
Answer: _____

4. Mixture Word Problems

A mixture blends two ingredients with different prices per kilogram. The same two equations return: one for the **total amount** mixed, one for the **total value**. Define each variable as a number of kilograms so the units in the value equation agree.

Example 5

A snack shop mixes cashews, which cost \$8/kg, with peanuts, which cost \$3/kg, to make a party mix. The mix weighs 20 kg in total and sells for \$110.

(a) Let c represent the number of kg of cashews used and p represent the number of kg of peanuts used. Write a system of two equations (one for the total weight, one for the total value), then solve it using **substitution**.

(b) State how many kg of each ingredient are in the mix, and check your answer.
Answer: _____

Lesson 8: Unit Review: Systems of Linear Equations

Every Unit 8 method on one page of summary tables, then mixed practice for the test.

1. The Three Solving Methods

A **solution** to a system is the ordered pair (x, y) that satisfies *both* equations at once. Three methods find it; all three must agree.

Method	When to use it	Watch out for
Graphing (L1)	Both equations are already $y = mx + b$, or rewrite easily; you want to *see* the answer.	A graph only estimates — it is unreliable unless the lines cross at integer coordinates.
Substitution (L3, L4)	One equation is solved for a variable, or some variable has coefficient 1 or -1 so you can isolate it cleanly.	Substitute into the *other* equation (using the same one gives $0 = 0$); back-substitute and check in *both* originals.
Elimination (L5, L6)	Both equations are in $Ax + By = C$ form and no coefficient is 1 or -1.	Opposite coefficients: *add*. Matching coefficients: *subtract* — which flips the sign of every term, the constant included. If nothing matches, multiply one or both equations first.

2. Classifying a System (Lesson 2)

Put both equations in slope-intercept form and compare m, then b.

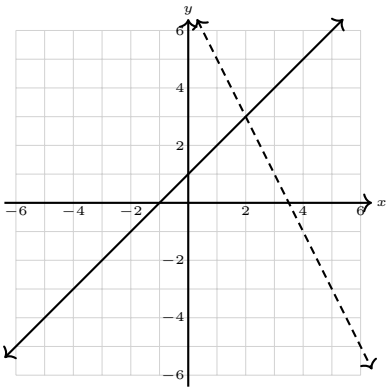
Outcome	Slopes / intercepts	Graph	Algebra ends with
One solution	Different slopes	Lines cross once	$x =$ a number
No solution	Same m, different b	Parallel lines	A false statement, e.g. $0 = 8$
Infinitely many	Same m, same b	One line drawn twice	A true statement, e.g. $0 = 0$

3. Word Problems (Lesson 7)

Name the two unknowns with variables, turn each sentence of given information into one equation, solve by whichever method above is quickest, then answer the question *in words*. For a rectangle, remember the perimeter: $P = 2L + 2W$.

Example 1 — Graphing

The system $y = x + 1$ and $y = -2x + 7$ is graphed below; the lines cross at integer coordinates.



(a) State the solution: $x =$_____, $y =$_____

(b) Write the solution as an ordered pair, and check it in **both** original equations.

Example 2 — Classifying

Without graphing, state whether each system has one solution, no solution, or infinitely many.

(a) $y = 6x - 1$, $y = 6x + 5$

(b) $y = 2x + 7$, $y = -4x + 1$

(c) $y = -3x + 6$, $y = -3x + 6$

(d) $y = x - 9$, $y = -x - 9$

Examples 3 and 4 — Substitution

Solve each system using substitution. In (4), isolate a variable first.

3. $y = 4x - 3$ $\quad 2x + y = 15$

4. $x + 3y = 19$ $\quad 2x - y = 3$

Examples 5 and 6 — Elimination

Solve each system using elimination. In (6), multiply each equation by a suitable constant to eliminate y.

5. $3x + 2y = 18 \qquad x - 2y = -2$

6. $4x + 3y = 23 \qquad 3x + 2y = 16$

Example 7 — Word Problem

A youth soccer tournament sold 150 tickets, some at the student price of \$6 and some at the adult price of \$10. Total ticket sales were \$1220.

(a) Let x be the number of student tickets and y the number of adult tickets. Write a system of two equations that models this situation.

(b) Solve the system to find the number of student tickets and the number of adult tickets sold.

Challenge 1 — Choosing a Method

Consider the system $\quad 5x - 3y = 19 \qquad 2x + y = 12.$

(a) Decide whether substitution or elimination is faster here, and explain your choice in one sentence.

(b) Solve the system using the method you chose.

Challenges 2 and 3 — Mixed Skills

2. A rectangular deck has a perimeter of 60 m. Its length is 6 m more than twice its width.

(a) Let L be the length and W the width. Write a system of two equations that models this situation.

(b) Solve the system to find the length and width of the deck.

3. Shop A charges a \$30 diagnostic fee plus \$45 per hour of labour; Shop B charges a \$55 diagnostic fee plus \$45 per hour of labour.

(a) Write an equation for the total cost at each shop in terms of the hours h of labour.

(b) Without solving, classify the system and explain using the slopes and intercepts.

(c) What does your answer to (b) mean about the two shops' pricing?